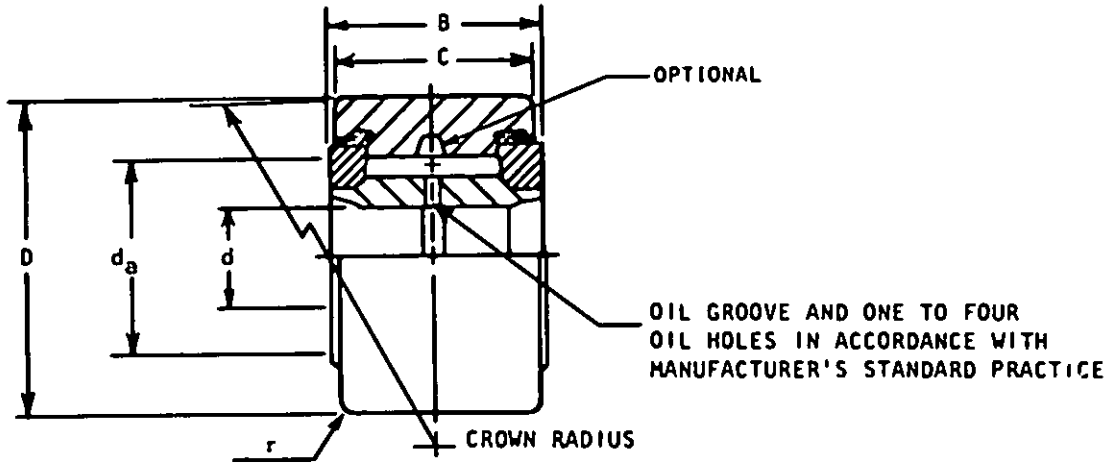
FED. SUP CLASS
3110




Dimensions in inches

| Dash No. | d Bore Diameter | D Outside Diameter | C Outer Ring Width | B Overall Width | $d_a$ Clamping Diameter Minimum | Crown Radius Ref | r Max Fillet Corner Will Clear | Total Radial Play Max (A) | Capacity As a Track Roller | Track Capacity lbf | Limit Load Rating lbf (A) | Mass (Approx) lbf |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| -121 | 0.2500 | 0.7500 | 0.500 | 0.5625 | 0.500 | 10.00 | 0.016 | 0.0027 | 2000 | 1030 | 2000 | 0.06 |
| -141 | 0.2500 | 0.8750 | 0.500 | 0.5625 | 0.500 | 10.00 | 0.016 | 0.0027 | 2000 | 1160 | 2600 | 0.08 |
| -161 | 0.3125 | 1.0000 | 0.625 | 0.6875 | 0.641 | 12.00 | 0.032 | 0.0027 | 2900 | 1400 | 3900 | 0.15 |
| -181 | 0.3125 | 1.1250 | 0.625 | 0.6875 | 0.641 | 12.00 | 0.032 | 0.0027 | 2900 | 1670 | 3900 | 0.17 |
| -201 | 0.3750 | 1.2500 | 0.750 | 0.8125 | 0.766 | 14.00 | 0.047 | 0.0027 | 4500 | 2330 | 5500 | 0.24 |
| -221 | 0.3750 | 1.3750 | 0.750 | 0.8125 | 0.766 | 14.00 | 0.047 | 0.0027 | 4500 | 2570 | 5500 | 0 30 |
| -241 | 0.4375 | 1.5000 | 0.875 | 0.9375 | 0.891 | 20.00 | 0.063 | 0.0027 | 5800 | 3300 | 7300 | 0.41 |
| -261 | 0.4375 | 1.6250 | 0.875 | 0.9375 | 0.891 | 20.00 | 0.063 | 0.0027 | 5800 | 3660 | 7300 | 0.50 |
| -281 | 0.5000 | 1.7500 | 1.000 | 1.0625 | 1.047 | 20.00 | 0.063 | 0.0027 | 8100 | 4500 | 10300 | 0.64 |
| -301 | 0.5000 | 1.8750 | 1.000 | 1.0625 | 1.047 | 20.00 | 0.063 | 0.0027 | 8100 | 4020 | 10300 | 0.80 |
| -321 | 0.6250 | 2.0000 | 1.250 | 1.3125 | 1.203 | 24.00 | 0.094 | 0.0027 | 11100 | 6550 | 13800 | 1.05 |
| -361 | 0.6250 | 2.2500 | 1.250 | 1.3125 | 1.203 | 24.00 | 0.094 | 0.0027 | 11100 | 7370 | 13800 | 1.32 |
| -401 | 0.7500 | 2.5000 | 1.500 | 1.5625 | 1.313 | 30.00 | 0.094 | 0.0027 | 16200 | 9250 | 21600 | 1.80 |
| -441 | 0.7500 | 2.7500 | 1.500 | 1.5625 | 1.313 | 30.00 | 0.094 | 0.0027 | 16200 | 10200 | 21600 | 2.25 |
| -481 | 1.0000 | 3.0000 | 1.750 | 1.8125 | 1.750 | 30.00 | 0.125 | 0.0027 | 24600 | 13400 | 32800 | 3.10 |
| -521 | 1.0000 | 3.2500 | 1.750 | 1.8125 | 1.750 | 30.00 | 0.125 | 0.0027 | 24600 | 14500 | 32800 | 3.62 |
| -561 | 1.1250 | 3.5000 | 2.000 | 2.0625 | 1.922 | 30.00 | 0.125 | 0.0027 | 31200 | 16300 | 41700 | 4.95 |
| -641 | 1.2500 | 4.0000 | 2.250 | 2.3125 | 2.281 | 30.00 | 0.125 | 0.0027 | 41000 | 20100 | 54700 | 7.85 |

User Activities: Army - AR, Navy - AS, Air Force - 11, DLA - CC

Review Activities: Army - AT, Navy - AS, Air Force - 11, 99

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new programming and design applications and for repetitive use shall be made from this document, when applicable.

APPROVED 8 Aug 80 REVISED (4) 31 Mar 81

| P.A. | INTERNATIONAL INTEREST | TITLE | MILITARY STANDARD |
|---|---|---|---|
| AIR FORCE-11<br>Other Cust<br>ARMY - AT<br>NAVY - AS | | BEARING, ROLLER, NEEDLE, TRACK ROLLER, SEALED, TYPE K, ANTIFRICTION, INCH | MS 21442 |
| PROCUREMENT SPECIFICATION MIL-B-3990 | | SUPERSEDES: | SHEET 1 OF 2 |

DD FORM 672-1

Project No. 3110-0676

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TABLE I: TOLERANCE LIMITS

Dimension in inches.

| d Basic Bore Diameter | | Allowable Deviation From d of Single mean Dia., dmp | | Allowable Deviation From Over-all Width B | |
|---|---|---|---|---|---|
| Over | Incl | High | Low | High | Low |
| 0.1875 | 0.7500 | +0.0002 | -0.0004 | +0.005 | -0.010 |
| 0.7500 | 1.2500 | +0.0001 | -0.0005 | +0.005 | -0.010 |

| D Basic Outside Diameter | | Allowable Deviation From D of Single Mean Dia., Dmp (A) | | Allowable Deviation From Outer Ring Width C | |
|---|---|---|---|---|---|
| Over | Incl | High | Low | High | Low |
| 0.625 | 4.000 | +0.0020 | -0.0005 | +0.005 | -0.005 |

NOTES:

(A) 1. MATERIAL: Steel, MIL-S-7493, MIL-S-8690, ASTM A304, A322, A331, QQ-S-700, FED. STD. 66, AISI/SAE Steel No. 50100, 51100, and 52100. Seals and backing rings - acetal resin per L-P-392 or nylon per L-P-410, type 6/6, wear resistant grade, elastomeric polyester. Construction optional.

2. FINISH:

A. Outer ring - outer ring chrome plated per QQ-C-320, Class 2. O.D. and O.D. corners .0004-.0007 in. thickness. faces minimum .0003 in thickness.

B. Inner ring - all external surfaces except the inner ring bore cadmium plated per QQ-P-416, Type I, Class 2. Alternate: all exposed surfaces have black oxide finish per MIL-C-13924.

C. Endplates - all exposed surfaces cadmium plated per QQ-P-416, Type I, Class 2.

3. LUBRICATION: MIL-G-81322 or MIL-G-23827. All bearings shall be prepacked with grease conforming to MIL-G-81322 unless otherwise specified. If MIL-G-23827 is required add the letter G after the MS part number.

4. MARKING: The marking shall consist of the MS part number and the manufacturer's identification in accordance with MIL-STD-130.

(A) 5. PART NUMBER: The MS part number shall consist of the MS number plus the dash number. Example: MS21442-121 or MS21442-121G.

6. PACKAGING: Bearings shall be individually packaged to the requirements of MIL-B-197. Package marked with manufacturer's name or trademark, and date of lubrication by month and year.

7. DIMENSIONS: All dimensions are after plating and are in inches.

8. TEMPERATURE: Bearing operating temperature range -65°F to 250°F.

9. DESIGN FEATURES: For design feature purposes, this standard takes precedence over documents referenced herein.

10. REFERENCED DOCUMENTS: Shall be of the issue in effect on date of invitations for bid.

(A) 11. LIMIT LOAD: The limit load rating listed can be defined as the maximum radial load that can be applied to a bearing without impairing the subsequent functioning of the bearing. The ultimate or static fracture load rating is not less than 1.5 times the limit load rating. The capacity as a track roller is the load the bearing will carry for an L-10 life of 20,000 revolutions.

User Activities: Army - Navy - Air Force - DLA -

Review Activities: Army - AT Navy - AS Air Force - 11,99 DLA - IS

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APPROVED 8 Aug 80 REVISED (A) 31 Mar 81

P.A. AIR FORCE-11 Other Cust ARMY - AT NAVY - AS

INTERNATIONAL INTEREST

TITLE: BEARING, ROLLER, NEEDLE, TRACK ROLLER, SEALED, TYPE X, ANTIFRICTION, INCH

MILITARY STANDARD

MS 21442

PROCUREMENT SPECIFICATION MIL-B-3990

SUPERSEDES:

SHEET 2 OF 2

DD FORM 672-1

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

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